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Machine learning-based analysis on factors influencing blood heavy metal concentrations in the Korean CHildren's ENvironmental health Study (Ko-CHENS)

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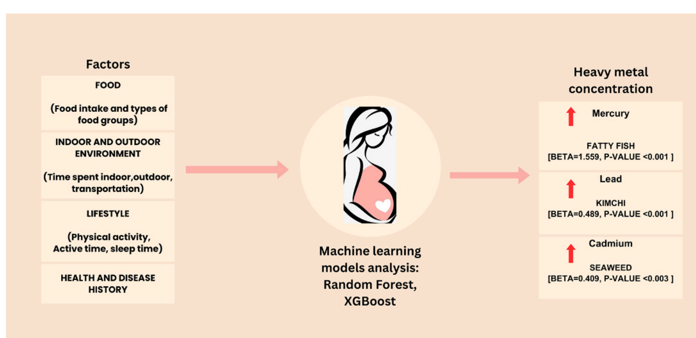
HIGHLIGHTS

- Machine learning models to predict factors influencing prenatal heavy metal exposure.
- Random forest and XGBoost models showed consistent results in early, late pregnancy and cord blood.
- Kimchi, seaweed, and fatty fish intake influenced lead, cadmium, and mercury exposure in Korean pregnant women.

GRAPHICAL ABSTRACT



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ABSTRACT

Heavy metal concentration in pregnant women affects neurocognitive and behavioral development of their infants and children. The majority of existing research focusing on pregnant women's heavy metal concentration has considered individual environmental factor. In this study, we aim to comprehensively consider lifestyle, food, and environmental factors to determine the most influential factor affecting heavy metal concentration in pregnant women. The Ko-CHENS (Korean CHildren health and ENvironmental Study) is a nationwide prospective birth cohort study in South Korea enrolling pregnant women from 2015 to 2020. A total of 5458 eligible pregnant women were included in this study, and 897 variables were included in questionnaire comprising: maternal general information, indoor and living environment, dietary habits, health behavior, exposure to chemicals. Lead, cadmium and mercury concentration on blood were measured in early, late pregnancy and in cord blood at birth. Variables that might be related to heavy metal concentrations were included in machine learning models. Random forest and XGBoost machine learning models were conducted for predictions. Both models had similar but better performance than multiple linear regression. Kimchi (β = 1.55), seaweed (β = 0.40), fatty fish (β = 1.55), intakes respectively affected lead, cadmium, and mercury exposure through early, late pregnancy and cord blood.

1. Introduction

Heavy metal exposure occurs through air, food, soil, and water, posing a serious threat to human health (Elder et al., 2015). Moreover, heavy metals are also known to accumulate in organs of the human body, such as the brain, heart, liver, and kidneys, disrupting normal biological functions (Fu and Xi, 2019), affecting the gastrointestinal, renal, and nervous systems, and causing skin lesions, vascular damage, immune system dysfunction, birth defects, and cancer (Balali-Mood and Sadeghi, 2021).

Pregnancy is a critical period affected by heavy metal exposure, as even low levels of heavy metal exposure during pregnancy can have adverse effects on the fetus and children. Heavy metals such as lead, mercury, and cadmium in the maternal blood can cross the placenta and accumulate in fetal tissues (Gundacker and Hengstschläger, 2012). The developmental origin of adult disease (DOHaD) hypothesis proposed by Barker et al. suggests that the early life environment can be associated with the onset of disease later in life (Mathew and Ayyar, 2012). Prenatal exposure to heavy metals can cause impaired cognitive development, affect birth outcomes, and hinder physical development even at very low concentrations (Al-Saleh et al., 2011).

Heavy metal exposure in pregnant women can occur through various routes. For instance, lead exposure can occur through the consumption of food such as vegetables, fruits, food storage cans, soil, dust, and drinking water. Exposure to cadmium can occur through contaminated food such as grain and cereal products, water, and cigarette smoke. Rice is an important source of cadmium exposure in Korea (Jeong et al., 2015). The sources of mercury, another highly toxic metal, include coal combustion, herbal medicines, skin creams, and the consumption of certain foods, particularly fish and other seafood (Kim et al., 2020a; Jyothi, 2020).

Given the various exposure sources of heavy metals and the potential for multiple exposures at one time, it is important to comprehensively evaluate and consider all exposure routes and sources. Traditional statistical approaches cannot effectively account for the interactions between the factors that influence heavy metal exposure, as well as the nonlinear relationship between exposures and heavy metal concentrations in high-dimensional data, resulting in low prediction performance. In contrast, machine learning (ML) models can identify patterns and correlations in datasets with high accuracy. Therefore, ML methods are gaining remarkable visibility in reproductive health, as they provide an efficient means to discover relationships in complex and nonlinear phenomena (Yaseen, 2021). Unlike traditional regression methods, which can be overfitted, the models obtained through ML methods are based on a cross-validation approach, making them highly robust. Therefore, our study sought to predict important parameters, such as the consumption of certain food items and lifestyle factors, affecting prenatal exposure to heavy metals (lead, mercury, and cadmium) in Korean pregnant women using different ML-based approaches.

2. Material and methods

2.1. Study design and population

The Korea Children's ENvironmental Health Study (Ko-CHENS) is a nationwide prospective birth cohort study that has been conducted in South Korea since May 2015. A detailed description of the study protocol for Ko-CHENS is provided in a previous study (Jeong et al., 2019). Here, we focused on the Ko-CHENS core cohort, which consists of 5458 pregnant women. From the total of 5458 pregnant women who completed the questionnaire, we excluded women who were foreigners, had multiple births, experienced miscarriages, abortions, chronic/pregnancy hypertension, diabetes, cardiovascular disease, intrauterine growth retardation, congenital malformations, or had extremely low (<500 kcal/day) or extremely high energy intake (>5000 kcal/day) (n = 724). Women who did not have a record of delivery outcomes were also excluded (n = 245). Of the remaining 4489 normal pregnant women, we excluded those with missing, non-response, and nonsensical data (n = 42). Finally, data from 4447 eligible pregnant women were used for analysis (Supplementary Fig. 1). For early pregnancy (before 20 weeks), blood 4447 eligible pregnant women were included; for late pregnancy (>28 weeks), data for 4265 eligible pregnant women were used; for cord blood, 3778 eligible pregnant women were included in our study (Supplementary Fig. 2). Written informed consent was obtained from the pregnant women. This study was approved by the Institutional Review Board of Ewha Womans University Hospital.

2.2. Characterization of lifestyle factors from questionnaire data

We considered all 897 variables from questionnaire data collected from women in early pregnancy. The 897 variables from questionnaire encompassed 13 categories: basic information, general characteristics, environmental factors, indoor environment, drug usage, allergies, mobile phone usage, radiation exposure, living habits, pregnancy-related information, diet, eating habits, and history of diseases. Detailed variables for each category are outlined in the Ko-CHENS cohort profile (Jeong et al., 2019).

Food frequency questionnaires (FFQ) were administered for the eating habits category. The frequency and amount of food consumption were surveyed for a total of 139 items. Nine levels were provided to indicate the frequency of eating: three or more times/day, two times/day, one time/day, 5–6 times/week, 3–4 times/week, 1–2 times/week, 2–3 times/month, one time/month, and rarely or never eating. For the consumption amount, three levels were assigned: small, medium, and large, corresponding to the standard intake of each food item. The frequency of food consumption was expressed as a daily unit for all items, while each level of food amount had different weights, multiplied by the standard intake of each food item: 0.5 for small, 1 for medium, and 1.5 for large. Subsequently, daily intakes (in grams) were calculated by multiplying the daily frequency and amount of food intake. Daily energy intakes (in kcal) were determined by multiplying the daily intakes by the standard calorie values. Finally, the 139 food items were categorized into 39 food groups based on their nutritional characteristics (Supplementary table 1), with each food group aggregating daily intakes and energy intakes.

2.3. Blood heavy metal concentration

Venous blood samples were collected from participants during early pregnancy (12–20 weeks), late pregnancy (>28 weeks) at outpatient visits, and at birth for cord blood. Vacuum blood collection tubes containing sodium ethylenediaminetetraacetic acid were utilized to collect whole blood samples (Vacutainer®, Beckton & Dickson, Franklin Lakes, NJ, USA). After storing the samples in a refrigerator, they were transferred to the laboratory for lead, mercury, and cadmium measurements. Blood metal levels (Pb, Cd) were quantified using a Perkin Elmer 900Z Graphite Furnace Atomic Absorption Spectrometry (GFAAs, Perkin Elmer, USA), Hg was analyzed using Gold amalgamation direct mercury analyzer (MA-3000, Nippon Instruments Corporation). The limit of detection (LOD) values for blood samples from pregnancy are 0.36 μg/dL for lead and 0.327 μg/L for cadmium. And cord blood was 0.179 μg/dL for lead and 0.184 μg/L for cadmium. Lead and cadmium concentrations were below the LOD values for early and late pregnancy and cord blood samples, with levels of 13.49 %, 14.97 %, and 8.31 % for lead, and 11.11 %, 6.65 %, and 32.49 % for cadmium in each period. Values below the LOD were imputed using maximum likelihood estimation methods (Chen et al., 2011). The mercury concentration did not include any values below the LOD value, indicating that all measurements were detectable and exhibited a log-normal distribution (Table 2). Using the mercury concentrations for each period, a likelihood function was constructed by multiplying the joint distribution of mercury and the target heavy metal in the same period and the conditional distribution of the target heavy metal concentration given mercury concentrations. The mean and variance for heavy metals in each period were estimated to maximize the likelihood function. Values below the LOD were replaced by random samples from the estimated distribution.

2.4. Statistical analysis

Descriptive characteristics of study participants were calculated using a *t*-test for continuous variables and a chi-square test for categorical variables. All blood heavy metal concentrations exhibited right-skewed distributions and were therefore log-transformed. The distribution of heavy metal concentrations was described in terms of geometric means and geometric standard deviations.

2.4.1. Data pre-processing

The accuracy of predictive models is highly dependent on the variables on which they are based. Variables that could be related to heavy metal concentration were thus selected for prediction through literature review, clinical knowledge, and statistical testing (Chowdhury and Turin, 2020). If a nominal variable had >6 levels, the levels were reduced based on their frequency. Ordinal variables were treated as continuous variables. Therefore, all continuous and ordinal variables were scaled using min-max scaling. Student's *t*-test was conducted for variables with two levels, while one-way analysis of variance (ANOVA) was performed for variables with three or more levels ($p < 0.1$). Pearson correlation was calculated for continuous variables. Variables with correlations exceeding the median of the overall absolute correlations in each period of heavy metal measurements were included. If a variable was determined to be relevant in any one period, it was used as a predictor for predicting the metal concentration. Based on these considerations, a total of 129, 121, and 114 variables were selected from the questionnaire for lead, cadmium, and mercury, respectively. Additionally, heavy metal concentrations from previous periods were included because they exhibited high correlations. Specifically, the concentration in early pregnancy was used as a predictor for late pregnancy concentration, and concentrations in both early and late pregnancy were used as predictors for cord blood concentration (Supplementary Fig. 1).

2.4.2. Modeling process

2.4.2.1. Machine learning model fitting. For ML, we randomly divided the data into a training set (80 %) for training models and a test set (20 %) for evaluating model performance (Supplementary Fig. 2). Two ML models, Random Forest (RF) (Breiman, 2001) and Extreme Gradient Boosting (XGBoost) (Chen and Guestrin, 2016), were employed to predict each heavy metal concentration. Hyperparameter tuning for the ML methods was optimized on the training dataset using five-fold cross-validation (Supplementary table 3).

The RF model is an ensemble decision tree model that randomly selects variables, trains subsets by bootstrapping, and repeatedly generates decision tree models. The predictions of each tree are then aggregated (Biau and Scornet, 2016). RF models can effectively detect interactions and nonlinear associations (Stafoggia et al., 2017). The variance of the responses is calculated as the variable importance score. Tuning parameters for the RF model included the number of trees (num.trees), the number of variables to possibly split at in each node (mtry), and the minimum node size to split at (min.node.size). The RF model was implemented using the 'ranger' R package (Wright and Ziegler, 2015).

The XGBoost model iteratively fits multiple decision tree models through different sampling methods. XGBoost employs the boosting algorithm, which improves model performance by using training errors as weights, thereby gradually optimizing the predictive performance of the model. Similar to RF, XGBoost also detects interactions and nonlinear associations. The variable importance score represents the relative contribution of each variable to the model, calculated by assessing each variable's contribution to each tree in the model. The tuning parameters for the XGBoost model included the maximum number of boosting rounds (nrounds), the maximum depth of the tree (max_depth), the minimum sum of instance weight needed in a child (min_child_weight), the subsample ratio of the training instances (subsample), the subsample ratio of columns when constructing each tree (colsample_bytree), and learning rate (eta) ["XGBoost Parameters – xgboost 1.5.0-dev documentation"]. The XGBoost model was implemented using the 'mlr' and 'xgboost' R packages (Bischl et al., 2016; Chen et al., 2015).

Multiple linear regression was also conducted to compare the predictive performance of the ML models. Stepwise variable selection was conducted based on the Akaike Information Criterion (AIC) to filter out irrelevant variables.

2.4.2.2. Model validation. Based on the root mean squared error (RMSE) and root mean absolute error (RMAE) of the log concentration of heavy metals, the performance of the three models was evaluated and compared in test set. RMSE represents the mean of the squared differences between the predicted and actual values, whereas RMAE represents the mean of the absolute differences. Lower RMSE and RMAE values indicate that the models predicted the heavy metal concentrations more accurately. Since the ML models randomly select variables and build trees, the model outcomes may vary in each training session. Therefore, we repeated the same process 50 times with fixed optimal parameters for each model. The average RMSE and RMAE values are summarized.

2.4.2.3. Variable importance. The variable importance scores from the

RF and XGBoost models were utilized to identify important variables for each heavy metal. The variable importance score in ML models indicates the degree to which a variable contributes to predicting a given outcome. Since we repeated the process 50 times for each model, we aggregated 50 sets of feature importance scores for each variable in each model. The importance scores from each repetition were scaled using min-max scaling methods prior to aggregation, and then averaged for each variable. As all categorical variables were transformed into dummy variables for analysis in the XGBoost model, dummy variables corresponding to the same categorical variable were aggregated to calculate the variable importance score for that specific categorical variable by averaging. Variable importance plots were generated for the top 10 variables with high variable importance scores, excluding confounders during variable selection.

2.4.2.4. Multiple linear regression for interpretation. Multiple linear regressions and simple linear regressions for the top 10 variables from each model were performed to address the limitation of interpretability in the machine learning results. The top 10 most important variables from the RF and XGBoost models were utilized, and linear regression analyses were conducted accordingly. First, simple (unadjusted) linear regression was carried out with log-transformed blood heavy metal concentrations in early and late pregnancy, as well as cord blood, as outcomes, and each of the top 10 variables as exposures. Next, multiple regression analysis was conducted, adjusting for key covariates including gestational age (in weeks), the mother's education (categorized as <high school, high school, university), monthly family income (categorized as <2000, 2000–4000, ≥4000 thousand won), job status for a year (Yes or No), the mother's age (categorized as <30, 30–34, ≥35 years), history of past diseases (Yes, No), parity (categorized as 1, >1), weight gain (in kg, calculated as weight after delivery – weight before pregnancy), and total energy intake (in kcal). The results from simple and multiple linear regression analyses were compared, and p-values < 0.05 were considered statistically significant. We performed a Linear Mixed Model (LMM) analysis to assess longitudinal changes in the effect of the most influential factor, adjusting for the same covariates used in the multiple linear regression models. The LMM analysis was conducted using SAS (version 9.4; SAS Institute Inc., Cary, NC, USA).

2.4.2.5. Shapley value. Interpreting ML model results solely through regression poses important limitations, as regression explains linear relationships only. To overcome these restrictions, explainable Artificial Intelligence (XAI) techniques have been recently developed (Adadi and Berrada, 2018). One such technique, the Shapley value, was utilized to ensure consistent interpretation and provide insights into the direction of associations (Shapley, 1953). The Shapley value represents the average contribution of interesting variables across all possible cases (Nohara et al., 2022). More importantly, this parameter can be readily implemented in machine learning algorithms (Lundberg and Lee, 2017). For each model of interest, Shapley dependence plots were employed to illustrate the relationship between variables and their impact on heavy metal concentration. The 'fastshap' R package was utilized for the RF model (Greenwell and Greenwell, 2020), whereas the 'SHAPforxgboost' package was employed for the XGBoost model (Liu and Just, 2020).

3. Results and discussion

3.1. Characteristics of the study population

The analyzed sample consisted of a total of 4447 women. Among them, 49.1 % were aged between 30 and 34. The average childbearing age of Korean women increased from 27 years in 1999 to 32 years in 2017, while the proportion of pregnancies among women aged 35 and older rose from 6.2 % in 1999 to 35.7 % in 2022 (Kim, 2023). Around, 44.4 % had a family income exceeding 4000 thousand won. Furthermore, 76.1 % were undergraduates, and 65.9 % held a job for at least a year. Many factors, such as educational aspirations and career goals, have been identified as important influences on childbearing age and fertility in Korea. Additionally, 87.3 % did not smoke, 76.9 % consumed alcohol, and 53.9 % had a history of disease. Moreover, 69.3 % were experiencing their first pregnancy. Notably, there were no significant differences in the proportion of women corresponding to the levels of each variable between the training and test sets (Table 1).

3.2. Prenatal heavy metal description

The median exposure levels for lead, cadmium, and mercury during early pregnancy were 0.66 μg/dL, 0.56 μg/L, and 2.06 μg/L, respectively (Table 2). In late pregnancy, the median exposure levels for lead, cadmium, and mercury were 0.66 μg/dL, 0.56 μg/L, and 2.05 μg/L, respectively. Additionally, in cord blood, the median exposure levels for lead, cadmium, and mercury were 0.61 μg/dL, 0.63 μg/L, and 1.72 μg/L, respectively. Interestingly, prenatal exposure to heavy metals in Korean pregnant women appears to be decreasing compared to previous years. For instance, a study by Jeong et al. (2017) reported cord blood exposure levels of 0.92 μg/dL for lead, 0.67 μg/dL for cadmium, and 5.10 μg/dL for mercury. This suggests a declining trend in heavy metal exposure among Korean women over time. Comparing these exposure levels to other populations, prenatal lead and cadmium exposure in Korean women seem to be lower than in the US population (Xu et al., 2021). The decrease in heavy metal exposure could be due to the phase-out of leaded gasoline and strict environmental regulations by the government, including those on consumer products, food processing, and waste management. Public health education, guided interventions, and national biomonitoring programs play a crucial role in reducing heavy metal exposure. Additionally, prenatal mercury exposure in Korea appears to be lower than previous reports from Japan but higher than those reported in the US and Canada (Lee et al., 2023). Seafood consumption in the Korean population, along with other environmental factors, could

Table 1
Characteristics of the study population at baseline.

	Overall (N = 4447) (%)	Train set (N = 3559) (%)	Test set (N = 888) (%)	P-value[a]
Age				0.193
<30	1214 (27.3)	976 (27.4)	238 (26.8)	
30–34	2183 (49.1)	1763 (49.5)	420 (47.3)	
≥35	1050 (23.6)	820 (23.0)	230 (25.9)	
Family income (thousand won)				0.5901
<200	278 (6.3)	229 (6.4)	49 (5.5)	
200–400	2194 (49.3)	1750 (49.2)	444 (50.0)	
≥400	1975 (44.4)	1580 (44.4)	395 (44.5)	
Education				0.3918
Under high school	517 (11.6)	415 (11.7)	102 (11.5)	
Undergraduate	3384 (76.1)	2719 (73.4)	665 (74.9)	
Over graduate	546 (12.3)	425 (11.9)	121 (13.6)	
Job status for a year				0.1206
Yes	2929 (65.9)	2324 (65.3)	605 (68.1)	
No	1518 (34.1)	1235 (34.7)	283 (31.9)	
Smoking history				0.3486
Yes	565 (12.7)	461 (13.0)	104 (11.7)	
No	3882 (87.3)	3098 (87.0)	784 (88.3)	
Drinking history				0.1254
Yes	1028 (23.1)	805 (22.6)	223 (25.1)	
No	3419 (76.9)	2754 (77.4)	665 (74.9)	
Diseases history				0.1432
Yes	2399 (53.9)	1900 (53.4)	499 (56.2)	
No	2048 (46.1)	1659 (46.6)	389 (43.8)	
Parity				0.3486
1	3080 (69.3)	2477 (69.6)	603 (67.9)	
More than 1	1367 (30.7)	1082 (30.4)	285 (32.1)	

[a] P-value for test of homogeneity between train and test data set.

Table 2
Blood heavy metal concentration in pregnant women.

Heavy metals	Period	N[a]	% < LOD	GM	GSD	Minimum	25th percentile	Median	75th percentile	Maximum
Lead (μg/dL)	Early pregnancy	4447	13.49	0.653	1.655	0.082	0.476	0.666	0.914	6.107
	Late pregnancy	4265	14.97	0.651	1.654	0.082	0.474	0.666	0.910	6.107
	Cord blood	3690	8.31	0.606	1.662	0.091	0.436	0.611	0.823	24.784
Cadmium (μg/L)	Early pregnancy	4447	11.11	0.568	1.561	0.122	0.424	0.569	0.760	3.533
	Late pregnancy	4265	6.65	0.567	1.555	0.122	0.424	0.569	0.757	3.533
	Cord blood	3690	32.49	0.646	1.530	0.114	0.491	0.663	0.860	3.705
Mercury (μg/L)	Early pregnancy	4447	0	2.106	1.596	0.141	1.554	2.061	2.822	12.019
	Late pregnancy	4265	0	2.102	1.598	0.141	1.548	2.054	2.824	12.019
	Cord blood	3690	0	1.756	1.571	0.059	1.312	1.727	2.313	11.505

LOD: limit of detection; GM: geometric mean; GSD: geometric standard deviation.
[a] Summary value are analyzed for pregnant women in study population.

explain the increased mercury concentration.

3.3. Comparison of model performance

The results of the prediction models exhibited both differences and similarities. ML models generally demonstrated similar or better performance compared to multiple regression models. Specifically, ML models performed slightly better when predicting lead concentration in late pregnancy and cord blood compared to the linear regression models. In early pregnancy, all three models showed similar performance. Regarding the prediction of cadmium concentration, ML models outperformed linear regression models across all periods. In the case of predicting mercury concentration, RF performed better in early pregnancy, whereas the multiple linear regression model performed better in late pregnancy and cord blood. Despite the slight differences in performance between linear and ML models, ML models accounted for nonlinear relationships and interactions among predictors, as well as computation time and sensitivity to outliers, providing a more comprehensive understanding (Table 3).

3.4. Feature importance

Various methods are employed to assess factors associated with heavy metal risk (Ghizadeh-Mehrjardi et al., 2021). However, there are no clear rules for factor selection (Guo et al., 2019). In this study, factors were selected based on different machine learning methods, further corroborated by literature review and available data (Zhang et al., 2011). Higher values indicate greater importance of factors for predicting prenatal heavy metal exposure. Variable importance plots for the top 10 variables for each model are presented in Figs. 1A, 1B, and 1C into the subsequent timeframe. Specifically, the influence of early pregnancy heavy metal exposure was considered in late pregnancy, and the corresponding early and late pregnancy exposures were considered when predicting cord blood exposure, as depicted in Figs. 1A, 1B, and 1C. In other words, our model assumed that previous timeframe exposure to heavy metals explains a significant portion of the current exposure. Therefore, it is important to consider other factors within the top 10 variables that could contribute to heavy metal exposure during early stages, thereby influencing heavy metal levels in later stages.

The most influential factor for lead exposure was kimchi intake. Kimchi is a traditional Korean dish of fermented vegetables, primarily made with Chinese cabbage and seasoned with chili pepper, garlic, ginger, and salted seafood (CAC, 2001). This result was consistently observed in both RF and XGBoost models in early and late pregnancy. Kimchi intake also had a slight effect on lead concentration in cord blood. The second most important factor explaining lead exposure was fruit intake, which ranked second in importance in RF model for both early, late pregnancy and among top 10 variable for cord blood lead exposure. Although it did not appear in the top 10 variables for the early pregnancy and cord blood in XGBoost model, fruit intake ranked fourth in importance in the late pregnancy model (Fig. 1A).

The most important factor associated with cadmium exposure was seaweed intake. This result was consistently identified in both RF and XGBoost models in early pregnancy. However, the impact of seaweed intake decreased markedly in late pregnancy and cord blood. The second most important factor was green/yellow vegetable intake in early and late pregnancy in the RF model, and milk intake in the XGBoost model. Green/yellow vegetable intake ranked ninth in importance in XGBoost, while tofu, soymilk intake ranked tenth in RF model for cord blood cadmium exposure (Fig. 1B).

Regarding mercury exposure, the most influential factor was fatty fish intake, which was consistent across both the RF and XGBoost models in early and late pregnancy. However, the effect of fatty fish intake decreased in late pregnancy and cord blood. The average time spent indoors was the second most important variable for the RF model, whereas lean fish intake ranked second for the XGBoost model. The average time spent indoors ranked third in importance for XGBoost, whereas lean fish intake did not appear in the top 10 most important variables for the RF model (Fig. 1C).

The differences in the top 10 variables between the RF and XGBoost models are influenced by algorithmic preferences inherent in the metrics used to calculate and interpret feature importance. RF model primarily employs mean decrease impurity (MDI) and mean decrease in accuracy (MDA) to identify significant features. By combining impurity-based and permutation-based methods, RF model provides a balanced perspective on feature importance (Polamuri, 2017).

Table 3
Prediction performance of multiple linear regression and machine learning models.

Heavy metals[b]	Performance	Early pregnancy			Late pregnancy			Cord blood		
		Linear regression	RF[a]	XGBoost	Linear regression	RF[a]	XGBoost[a]	Linear regression	RF[a]	XGBoost[a]
Lead	RMSE[a]	0.4856	0.4451	0.4890	0.4498	0.4451	0.4452	0.4202	0.4145	0.4151
	RMAE[a]	0.6181	0.6236	0.6207	0.5899	0.5873	0.5875	0.5606	0.5559	0.5560
Cadmium	RMSE[a]	0.4321	0.4305	0.4308	0.3732	0.3685	0.3668	0.3404	0.3250	0.3272
	RMAE[a]	0.5817	0.5814	0.5802	0.5322	0.5291	0.5261	0.5152	0.5035	0.5058
Mercury	RMSE[a]	0.4418	0.4407	0.4412	0.3100	0.3146	0.3159	0.2284	0.2350	0.2338
	RMAE[a]	0.5801	0.5782	0.5788	0.4923	0.4965	0.4980	0.4201	0.4268	0.4267

[a] RF: random forest; XGBoost: extreme gradient boosting; RMSE: root mean squared error; RMAE: root mean absolute error.
[b] Log-transformed heavy metal concentration.

(a)

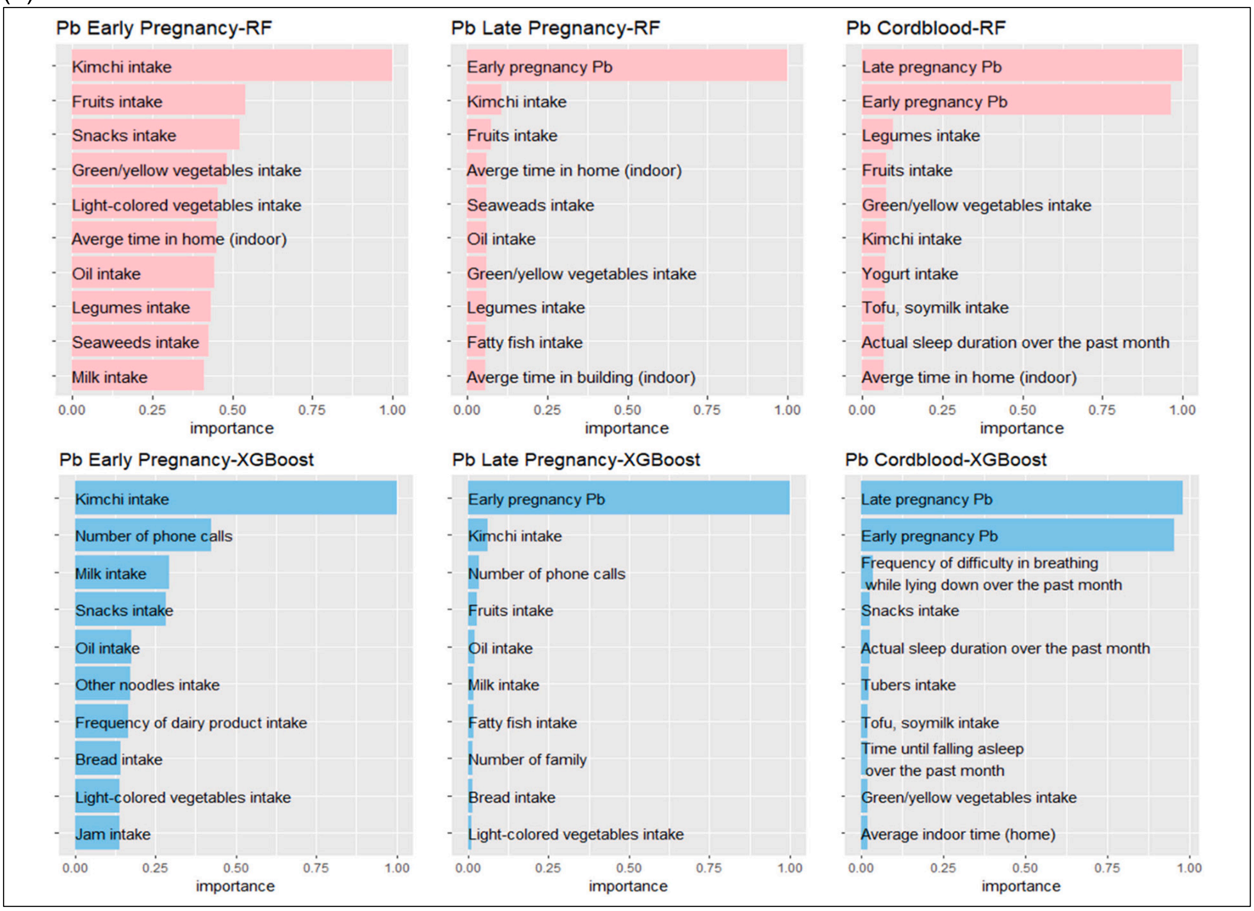

Fig. 1A. Top 10 influential factors and feature importance of prediction model affecting prenatal blood lead concentration.
Pb: lead; RF: random forest; XGBoost: extreme gradient boosting; Kimchi: Kimchi is a traditional Korean dish of fermented vegetables, primarily made with Chinese cabbage and seasoned with chili pepper, garlic, ginger, and salted seafood.

In contrast, XGBoost model evaluates feature importance using metrics such as gain, frequency, and cover. Gain is prioritized for tasks focusing on accuracy, while frequency and cover provide additional insights into feature usage and impact (Morde and Setty, 2019).

Additionally, differences in feature importance when analyzing early, late-pregnancy, and cord blood heavy metal exposure may stem from changes in pregnant women's behaviors, including dietary and cooking habits, which reflect their daily routines. These behaviors are likely influenced by environmental and psychological factors (Herzog-Petropaki et al., 2022).

3.5. Association between heavy metals and important factors

In our study, kimchi intake exhibited a positive and significant relationship with lead concentration in both early ($\beta = 0.489$, p-value < 0.001) and late pregnancy ($\beta = 0.381$, p-value < 0.001). However, kimchi intake was not significantly associated with cord blood lead concentration, possibly due to its lower importance in the prediction model and the discrepancy in coefficient sign observed in multiple regression analysis ($\beta_{multiple} = -0.048$, p-value $= 0.611$; $\beta_{single} = 0.375$, p-value < 0.001). Conversely, fruit intake showed a negative relationship with lead concentration in both early ($\beta = -0.032$, p-value $= 0.721$) and late pregnancy ($\beta = -0.066$, p-value $= 0.427$), although it was not statistically significant (Table 4 and Supplementary Table 4). Kimchi is a major source of lead exposure in the Korean population due to its high consumption (Lee et al., 2006). Notably, a study representing the Korean population found that vegetables had higher lead concentrations, potentially explaining the positive association between kimchi intake and lead levels in pregnant women (Park and Lee, 2013; Hong et al., 2020). This increased lead content in vegetables may result from soil contamination (Kim et al., 2020b).

Our results show seaweed intake had a positive and significant relationship with cadmium concentration in early pregnancy ($\beta = 0.409$, p-value $= 0.003$). However, green/yellow vegetable intake showed varying associations with cadmium concentration, being positive in early pregnancy ($\beta = 0.004$, p-value $= 0.982$) and cord blood ($\beta = 0.183$, p-value $= 0.197$), but negative in late pregnancy ($\beta = -0.014$, p-value $= 0.919$), although these coefficients were not statistically significant. Conversely, milk intake exhibited a negative and significant association with cadmium concentration in early pregnancy ($\beta = -0.305$, p-value $= 0.001$; Table 4 and Supplementary Table 5). Seaweed and green vegetables, like spinach, are considered major contributors to total cadmium intake in the Korean population (Lee et al., 2006). These findings are supported by a study by Park and Lee, which found that seaweed consumption was associated with increased blood cadmium levels, whereas milk consumption did not affect blood cadmium concentration (Park and Lee, 2013).We found fatty fish intake demonstrated a positive and mostly significant relationship with mercury concentration across all periods ($\beta_{early} = 1.559$, p-value < 0.001; $\beta_{late} = 0.531$, p-value < 0.001; $\beta_{cord} = 0.151$, p-value $= 0.092$). Conversely, average time spent at home exhibited a negative and significant relationship with mercury concentration in early pregnancy ($\beta = -0.160$, p-value $= 0.002$; Table 4 and Supplementary Table 6). Fatty fish intake has been an important source of mercury exposure in the Korean population due to its higher

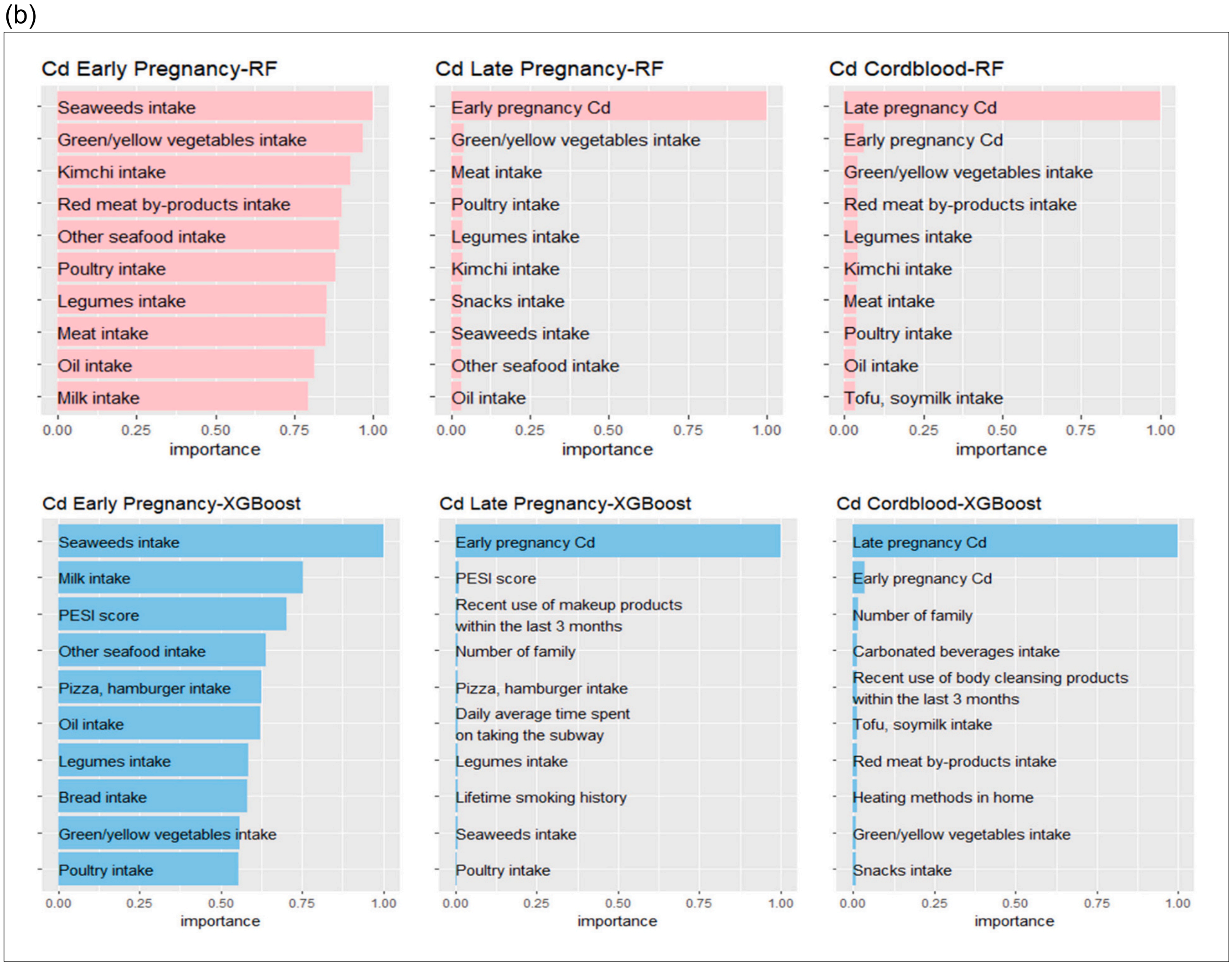

Fig. 1B. Top 10 influential factors and feature importance of prediction model affecting prenatal blood cadmium concentration.
Cd: cadmium; RF: random forest; XGBoost: extreme gradient boosting; Kimchi: Kimchi is a traditional Korean dish of fermented vegetables, primarily made with Chinese cabbage and seasoned with chili pepper, garlic, ginger, and salted seafood.

consumption. This association may be explained by the higher mercury concentrations per gram of body weight found in fatty fish species such as Spanish mackerel and Korean rockfish, as reported by Kim et al. (2020b).

Shapley values for kimchi intake for lead, seaweed intake for cadmium, and fatty fish intake for mercury increased as these factors increased in the early pregnancy model for both RF and XGBoost models evaluated (Supplementary Fig. 3A, B, and C). Across all metals in early pregnancy, the Shapley values for the most important variables increased as the values of factors increased, although this increasing trend was attenuated in late pregnancy. In cord blood, Shapley values for variables remained unchanged, as previous periods of heavy metal exposure strongly influenced cord blood concentration, confirming the association between the important predictors identified in our ML model analyses and the corresponding heavy metal concentration.

We examined whether the impact of each key dietary factor on blood heavy metal concentrations varied across early pregnancy, late pregnancy, and at birth (Supplementary table 8). For kimchi intake and lead (p-value = 0.10), as well as fatty fish intake and mercury (p-value = 0.69), the time-specific differences were not statistically significant. This may be because the most influential variables for lead and mercury were consistently significant across all time points in the prediction models, making time-dependent differences less apparent. However, the beta estimates for the interaction terms still changed over time, suggesting that the effects of these factors may slightly vary depending on the timing of exposure. In contrast, for seaweed intake and cadmium, a significant interaction with time was observed (p-value = 0.02), indicating that its effect on cadmium levels differed by pregnancy period. These findings suggest that the influence of certain lifestyle factors on blood heavy metal concentrations may depend on the timing of exposure during pregnancy.

3.6. Strengths and limitations

By employing various ML methods, we aimed to predict prenatal heavy metal exposure within a birth cohort study representing Korean women nationwide. This approach allowed us to overcome the limitations of traditional techniques and enhance prediction accuracy. The temporal sequence is critical in developing a prediction model. In our study, lifestyle and dietary information was collected at enrollment, reflecting behaviors during pregnancy period. The half-lives of heavy metals in blood vary: approximately 50 days for mercury, 3–4 months for the fast component of cadmium and 1–2 months for lead (McClam et al., 2023). Blood heavy metal concentrations reflect both recent and chronic exposures, though acute exposures can also influence heavy metal levels. Given these dynamics, lifestyle factors recorded at enrollment likely impact heavy metal exposure throughout pregnancy. Our findings highlight the need for interventions before conception and during pregnancy to reduce maternal heavy metal exposure and associated risks. Furthermore, dividing the dataset into training and test sets helped mitigate overfitting issues commonly encountered in validation methods. The ML methods employed herein effectively handled the

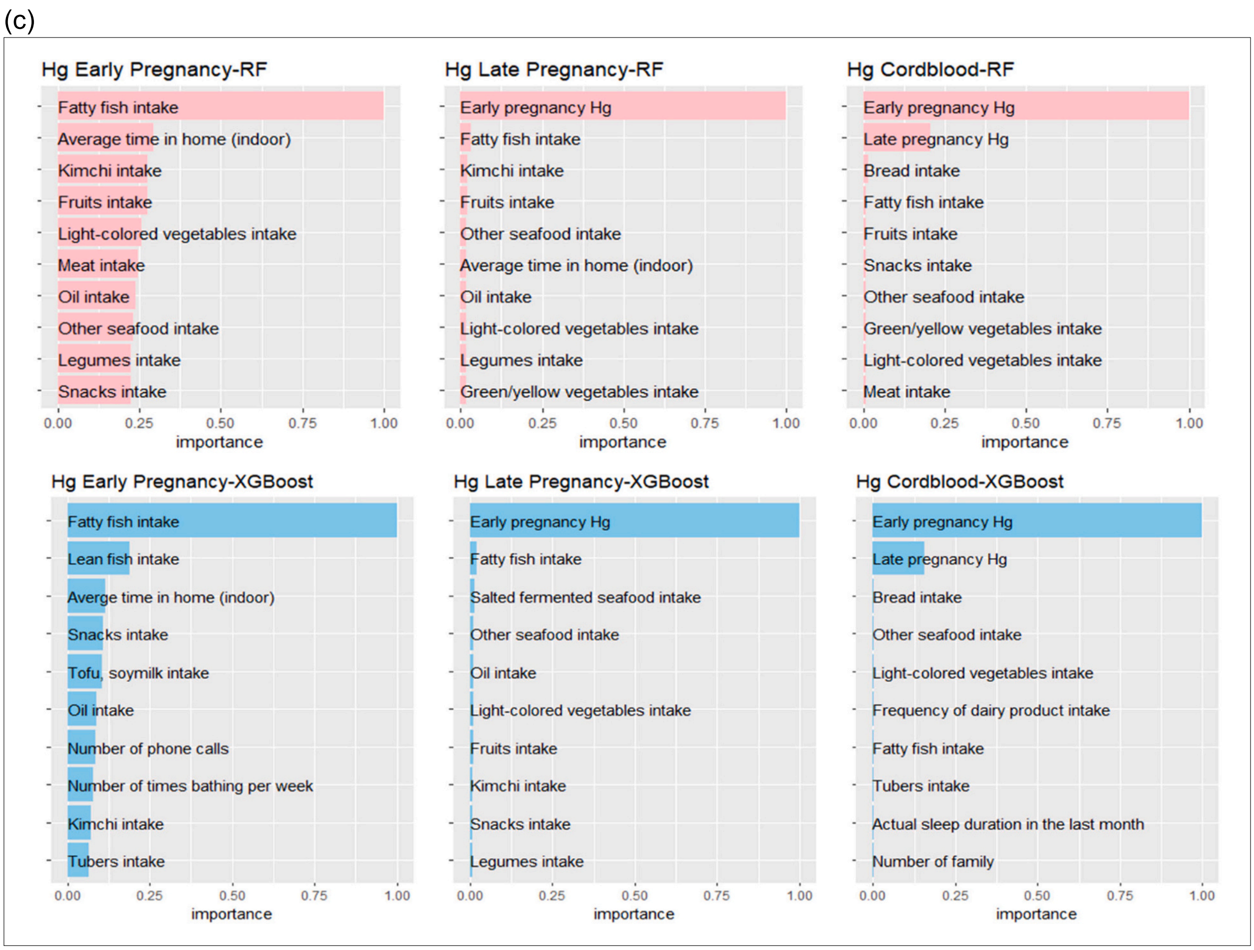

Fig. 1C. Top 10 influential factors and feature importance of prediction model affecting prenatal blood mercury concentration.
Hg: mercury; RF: random forest; XGBoost: extreme gradient boosting; Kimchi: Kimchi is a traditional Korean dish of fermented vegetables, primarily made with Chinese cabbage and seasoned with chili pepper, garlic, ginger, and salted seafood.

complexity and hidden relationships among the factors included in the prediction model. Previous studies have primarily focused on heavy metal exposure through environmental sources such as air, water, and soil. In contrast, our study takes a more comprehensive approach by examining the influence of various demographic characteristics, lifestyle factors, environmental conditions, indoor environments, and living habits on heavy metal exposure among Korean pregnant women. By integrating these multifaceted factors, our research provides a more detailed understanding of the diverse contributors to prenatal heavy metal exposure.

Our results demonstrated that mobile phone usage was a contributing factor to prenatal lead and mercury concentrations. Although research on the direct impact between mobile phone use and heavy metal exposure is limited, one possible mechanism involves radiofrequency radiation (RFR). RFR exposure may facilitate energy transfer that increases blood-brain barrier (BBB) permeability to macromolecules (Choi et al., 2017). This increased permeability could amplify the neurotoxic effects of heavy metals. These findings underscore the need for further investigation into the role of mobile phone usage in heavy metal exposure, particularly during critical periods like pregnancy.

Our study highlights dietary sources of heavy metals using advanced machine learning methods, offering a novel approach to refine exposure risk assessments. These results enable the prioritization of specific interventions by identifying critical dietary contributors to heavy metal exposure. Moreover, our findings provide a more precise understanding of population-specific dietary patterns, facilitating the development of targeted public health strategies for effective mitigation and risk reduction. To the best of our knowledge, this is the first epidemiological study to identify important factors influencing prenatal heavy metal exposure in Korean women.

However, we encountered noteworthy limitations when building and implementing the ML models. ML methods are highly efficient at extracting information but are subject to limitations such as the number, type, and quality of data available, which can significantly impact model performance (Qian et al., 2014). Additionally, exposure assessment relied on the responses of the participants to the questionnaire, introducing the possibility of recall bias. Similarly, the selection of participants based on questionnaire responses could also introduce selection bias. While we included various sources of heavy metal exposure in the ML model, some sources such as water and soil, which may influence heavy metal exposure, could not be included due to data availability constraints. We selected variables using statistical approaches, and simple regression and chi-square tests were conducted without considering potential confounding variables. As a result, variables unrelated to the outcome may have been included, potentially affecting the relationships between key variables and introducing bias (Pearce et al., 2007). Moreover, the generalizability of our results is limited, as all study participants were Korean women.

4. Conclusion

Our study utilized machine learning models to predict and validate the crucial lifestyle factors impacting heavy metal exposure in pregnant women. We discovered that kimchi intake influenced lead exposure,

Table 4
Pregnant women factors associated with prenatal heavy metal exposure.

Heavy metal	Sr. no.[c]	Early pregnancy[a]			Late pregnancy[a]			Cord blood[a]		
		Exposure factors	β estimate	SE[d]	Exposure factors	β estimate	SE[d]	Exposure factors	β estimate	SE[d]
Lead	1.	Kimchi[b] intake	0.489*	0.095	Kimchi[b] intake	0.381*	0.089	Frequency of difficulty breathing while lying down in the past month	0.024	0.032
	2.	Number of phone calls	0.885*	0.225	Number of phone calls	0.441	0.209	Snacks intake	−0.492	0.153
	3.	Milk intake	0.004	0.112	Fruits intake	−0.066	0.083	Actual sleep duration in the past month	−0.159	0.061
	4.	Snacks intake	−0.447	0.158	Oil intake	−0.276	0.086	Tubers intake	0.104	0.133
	5.	Oil intake	−0.287	0.094	Milk intake	−0.162	0.098	Tofu, soymilk intake	−0.035	0.103
	6.	Other Noodles intake	0.291	0.131	Fatty fish intake	0.312	0.160	Time until falling asleep over the past month	0.047	0.073
	7.	Frequency of dairy product intake	−0.143	0.038	Number of family	0.115	0.062	Green/yellow vegetables intake	0.167	0.172
	8.	Bread intake	−0.201	0.115	Bread intake	−0.049	0.102	Average time in home (indoor)	−0.047	0.055
	9.	Light-colored Vegetables intake	0.346	0.162	Light-colored vegetables intake	0.253	0.150	Legumes intake	0.131	0.100
	10.	Jam intake	0.235	0.194	Average time in home (indoor)	−0.029	0.061	Fruits intake	0.105	0.086
	11.	Fruits intake	−0.032	0.089	Seaweeds intake	0.257	0.146	Kimchi[b] intake	−0.048	0.094
	12.	Green/Yellow Vegetables intake	−0.158	0.209	Green/yellow vegetables intake	−0.171	0.195	Yogurt intake	0.003	0.087
	13.	Average time in home (indoor)	−0.019	0.057	Legumes intake	−0.245	0.099	–	–	–
	14.	Legumes intake	−0.275	0.107	Average time in building (indoor)	−0.048	0.101	–	–	–
	15.	Seaweeds intake	0.270	0.158	–	–	–	–	–	–
Cadmium	1.	Seaweeds intake	0.409*	0.137	PESI score	−0.106	0.054	Number of family	−0.060	0.051
	2.	Milk intake	−0.305*	0.090	Use of makeup products within the last 3 months	−0.031	0.017	Carbonated beverages intake	−0.068	0.110
	3.	PESI score	−0.146	0.063	Number of family members	−0.079	0.050	Use of body cleansers within the last 3 months	0.058*	0.027
	4.	Other seafood intake	0.129	0.105	Pizza, hamburger intake	−0.088	0.065	Tofu, soymilk intake	0.177*	0.082
	5.	Pizza, hamburger intake	−0.183	0.076	Daily average time spent on taking the public transportation	−0.205	0.075	Red meat by-products intake	−0.060	0.067
	6.	Oil intake	−0.126	0.082	Legumes intake	−0.064	0.080	Heating method in the house: individual	−0.088	0.013
	7.	Legumes intake	−0.145	0.093	Lifetime smoking history:Yes	0.067	0.017	Heating method in the house: none	0.010	0.132
	8.	Bread intake	−0.203	0.095	Seaweeds intake	0.183	0.116	Green/yellow vegetables intake	0.183*	0.142
	9.	Green/yellow vegetables intake	0.004*	0.157	Poultry intake	−0.049	0.077	Snacks intake	−0.225	0.125
	10.	Poultry intake	−0.097	0.090	Green/yellow vegetables intake	−0.014	0.135	Legumes intake	−0.054	0.079
	11.	Kimchi[b] intake	0.197	0.083	Meat intake	0.056	0.099	Kimchi[b] intake	0.097	0.075
	12.	Red meat by-products intake	−0.120	0.075	Kimchi[b] intake	0.072	0.072	Meat intake	0.034	0.108
	13.	Meat intake	0.013	0.123	Snacks intake	0.135	0.119	Poultry intake	−0.033	0.078
	14.	–	–	–	Other seafood intake	0.125	0.089	Oil intake	−0.062	0.072
	15.	–	–	–	Oil intake	−0.067	0.071			
Mercury	1.	Fatty fish intake	1.559*	0.170	Fatty fish intake	0.531*	0.116	Bread intake	−0.017	0.057
	2.	Lean fish intake	0.580	0.143	Salted fermented seafood intake	0.075	0.117	Other seafood intake	−0.063	0.061
	3.	Average time in home (indoor)	−0.160	0.052	Other seafood intake	0.198	0.079	Light-colored vegetables intake	0.087	0.081
	4.	Snacks intake	−0.572	0.145	Oil intake	−0.137	0.063	Frequency of dairy product intake	0.024	0.018
	5.	Tofu, soymilk intake	−0.274	0.097	Light-colored vegetables intake	−0.174	0.106	Fatty fish intake	0.151	0.090
	6.	Oil intake	−0.201	0.085	Fruits intake	0.079	0.058	Tubers intake	0.090	0.071
	7.	Number of phone calls	0.698	0.206	Kimchi[b] intake	0.095	0.064	Actual sleep duration in the past month	−0.108	0.032
	8.	Number of times bathing per week	0.109	0.025	Snacks intake	−0.121	0.105	Number of family	0.063	0.035
	9.	Kimchi[b] intake	0.035	0.087	Legumes intake	−0.081	0.071	Fruits intake	0.030	0.046
	10.	Tubers intake	−0.410	0.126	Average time in home (indoor)	−0.079	0.037	Snacks intake	0.061	0.083
	11.	Fruits intake	0.168	0.080	Green/yellow vegetables intake	0.205	0.139	Green/yellow vegetables intake	−0.046	0.112

(continued on next page)

Table 4 (*continued*)

Heavy metal	Sr. no.[c]	Early pregnancy[a]			Late pregnancy[a]			Cord blood[a]		
		Exposure factors	β estimate	SE[d]	Exposure factors	β estimate	SE[d]	Exposure factors	β estimate	SE[d]
	12.	Light-colored vegetables intake	−0.044	0.125	–	–	–	Meat intake	0.156	0.067
	13.	Meat intake	−0.110	0.116	–	–	–	–	–	–
	14.	Other seafood intake	0.418	0.109	–	–	–	–	–	–
	15.	Legumes intake	−0.179	0.098	–	–	–	–	–	–

[a] Multiple linear models adjusted for the model: mother's gestational age, education level, family income, job status for a year, age, history of past diseases, parity, pregnancy weight gain, total energy intake.
[b] Kimchi: Kimchi is a traditional Korean dish of fermented vegetables, primarily made with Chinese cabbage and seasoned with chili pepper, garlic, ginger, and salted seafood.
[c] Sr.no.: serial number from machine learning model.
[d] SE: standard error.
[*] p-value < 0.05

seaweed intake affected cadmium exposure, and fatty fish intake increased mercury exposure in Korean pregnant women. However, additional studies are recommended to validate our findings regarding the lifestyle factors influencing prenatal heavy metal exposure.

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Declaration of competing interest

The authors declare that have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2025.179401.

Data availability

Data are provided from the National Institute of Environmental Research (NIER). The authors have no right to share or provide the data.

References

Adadi, A., Berrada, M., 2018. Peeking inside the black-box: a survey on explainable artificial intelligence (XAI). IEEE Access 6, 52138–52160. https://doi.org/10.1109/ACCESS.2018.2870052.

Al-Saleh, I., Shinwari, N., Mashhour, A., Mohamed, G.E., Rabah, A., 2011. Heavy metals (lead, cadmium and mercury) in maternal, cord blood and placenta of healthy women. Int. J. Hyg. Environ. Health 214, 79–101. https://doi.org/10.1016/j.ijheh.2010.10.001.

Balali-Mood, M., Sadeghi, M., 2021. Toxic mechanisms of five heavy metals: mercury, lead, chromium, cadmium, and arsenic. Front. Pharmacol. 12, 643972. https://doi.org/10.3389/fphar.2021.643972.

Biau, G., Scornet, E., 2016. A random forest guided tour. Test 25, 197–227. https://doi.org/10.1007/s11749-016-0481-7.

Bischl, B., Lang, M., Kotthoff, L., Schiffner, J., Richter, J., Studerus, E., Jones, Z.M., 2016. mlr: machine learning in R. J. Mach. Learn. Res. 17, 5938–5942. http://jmlr.org/papers/v17/15-066.html. (Accessed 20 August 2024).

Breiman, L., 2001. Random forests. Mach. Learn. 45, 5–32. https://doi.org/10.1023/A:1010933404324.

CAC, 2001. Codex Standard for Kimchi (CODEX STAN 223-2001). Codex Ali mentarius Commission, Rome, Italy.

Chen, T., Guestrin, C., 2016. XGBoost: a scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. ACM, pp. 785–794. https://doi.org/10.1145/2939672.2939785.

Chen, H., Quandt, S.A., Grzywacz, J.G., Arcury, T.A., 2011. A distribution-based multiple imputation method for handling bivariate pesticide data with values below the limit of detection. Environ. Health Perspect. 119, 351–356. https://doi.org/10.1289/ehp.1002124.

Chen, T., He, T., Benesty, M., Khotilovich, V., Tang, Y., Cho, H., Zhou, T., 2015. Xgboost: Extreme Gradient Boosting. R Package Version 0.4-2. R J. 4, 1–4. https://doi.org/10.32614/CRAN.package.xgboost.

Choi, K.H., Ha, M., Ha, E.H., Park, H., Kim, Y., Hong, Y.C., Lee, A.K., Hwa Kwon, J., Choi, H.D., Kim, N., Kim, S., Park, C., Jul 2017. Neurodevelopment for the first three years following prenatal mobile phone use, radio frequency radiation and lead exposure. Environ. Res. 156, 810–817. https://doi.org/10.1016/j.envres.2017.04.029 (Epub 2017 May 13. PMID: 28511138).

Chowdhury, M.Z.I., Turin, T.C., 2020. Variable selection strategies and its importance in clinical prediction modelling. Fam. Med. Community Health 8, e000262. https://doi.org/10.1136/fmch-2019-000262.

Elder, A., Nordberg, G.F., Kleinman, M., 2015. Routes of exposure, dose, and toxicokinetics of metals. In: Handbook on the Toxicology of Metals. Academic Press, pp. 45–74.

Fu, Z., Xi, S., 2019. The effects of heavy metals on human metabolism. Toxicol. Mech. Methods 29, 653–659. https://doi.org/10.1080/15376516.2019.1701594.

Ghizadeh-Mehrjardi, R., Fathizad, H., Ali Hakimzadeh Ardakani, M., Davatgar, N., Malekian, A., 2021. Spatio-temporal analysis of heavy metals in arid soils at the catchment scale using digital soil assessment and a random forest model. Remote Sens. 13, 1698. https://doi.org/10.3390/rs13091698.

Greenwell, B., Greenwell, M.B., 2020. Package 'fastshap', The Comprehensive R Archive Network. https://cran.r-project.org/web/packages/fastshap/index.html. (Accessed 20 August 2024).

Gundacker, C., Hengstschläger, M., 2012. The role of the placenta in fetal exposure to heavy metals. Wien. Med. Wochenschr. 162, 201–206. https://doi.org/10.1007/s10354-012-0074-3.

Guo, Z., Adhikari, K., Chellasamy, M., Liu, D., 2019. Selection of terrain attributes and its scale dependency on soil organic carbon prediction. Geoderma 340, 303–312. https://doi.org/10.1016/j.geoderma.2019.01.023.

Herzog-Petropaki, N., Derksen, C., Lippke, S., 2022. Health behaviors and behavior change during pregnancy: theory-based investigation of predictors and interrelations. Sexes 3 (3), 351–366.

Hong, Y.K., Kim, J.W., Lee, S.P., et al., 2020. Heavy metal remediation in soil with chemical amendments and its impact on activity of antioxidant enzymes in lettuce (*Lactuca sativa*) and soil enzymes. Appl. Biol. Chem. 63, 42. https://doi.org/10.1186/s13765-020-00526-w.

Jeong, K.S., Park, H., Ha, E., et al., 2015. Performance IQ in children is associated with blood cadmium concentration in early pregnancy. J. Trace Elem. Med. Biol. 30, 107–111. https://doi.org/10.1016/j.jtemb.2014.11.007.

Jeong, K.S., Ha, E., Shin, J.Y., et al., 2017. Blood heavy metal concentrations in pregnant Korean women and their children up to age 5 years: Mothers' and Children's Environmental Health (MOCEH) birth cohort study. Sci. Total Environ. 605, 784–791.

Jeong, K.S., Kim, S., Kim, W.J., et al., 2019. Cohort profile: beyond birth cohort study - the Korean Children's Environmental Health study (Ko-CHENS). Environ. Res. 172, 358–366. https://doi.org/10.1016/j.envres.2018.12.009.

Jyothi, N.R., 2020. Heavy metal sources and their effects on human health. In: Rai, M., Acharya, D., Govindaraju, K. (Eds.), Heavy Metals-Their Environmental Impacts and Mitigation. Springer Nature, pp. 115–132. https://doi.org/10.5772/intechopen.91574.

Kim, T., 2023. The impact of working hours on pregnancy intention in childbearing-age women in Korea, the country with the world's lowest fertility rate. PLoS One 18 (7), e0288697. https://doi.org/10.1371/journal.pone.0288697.

Kim, B., Shah, S., Park, H.S., et al., 2020a. Adverse effects of prenatal mercury exposure on neurodevelopment during the first 3 years of life modified by early growth velocity and prenatal maternal folate level. Environ. Res. 191, 109909. https://doi.org/10.1016/j.envres.2020.109909.

Kim, T.H., Kim, J.H., Kim, Le, et al., 2020b. Exposure assessment and safe intake guidelines for heavy metals in consumed fishery products in the Republic of Korea. Environ. Sci. Pollut. Res. 27, 33042–33051. https://doi.org/10.1007/s11356-020-09624-0.

Lee, H.S., Cho, Y.H., Park, S.O., Kye, S.H., Kim, B.H., Hahm, T.S., Kim, M., Lee, J.O., Kim, C.I., 2006. Dietary exposure of the Korean population to arsenic, cadmium, lead and mercury. J. Food Compos. Anal. 19, S31–S37.

Lee, S., Kim, J.H., Moon, H.B., et al., 2023. Effects of mercury exposure on fetal body burden and its association with infant growth. Environ. Res. 217, 114780. https://doi.org/10.1016/j.envres.2022.114780.

Liu, Y., Just, A., 2020. SHAPforxgboost: SHAP plots for 'XGBoost'. R package version 0.1.0. https://github.com/liuyanguu/SHAPforxgboost. (Accessed 20 August 2024).

Lundberg, S.M., Lee, S.I., 2017. A unified approach to interpreting model predictions. In: Advances in Neural Information Processing Systems. Curran Associates Inc, pp. 4765–4774. https://doi.org/10.48550/arXiv.1705.07874.

Mathew, V., Ayyar, S.V., 2012. Developmental origins of adult diseases. Indian J. Endocrinol. Metab. 16, 532–541. https://doi.org/10.4103/2230-8210.98005.

McClam, M., Liu, J., Fan, Y., Zhan, T., Zhang, Q., Porter, D.E., Scott, G.I., Xiao, S., Aug 25, 2023. Associations between exposure to cadmium, lead, mercury and mixtures and women's infertility and long-term amenorrhea. Arch. Public Health 81 (1), 161. https://doi.org/10.1186/s13690-023-01172-6.

Morde, V., Setty, V.A., 2019. XGBoost algorithm: long may she reign. In: Towards Data Science (April 8, 2019. 2019).

Nohara, Y., Matsumoto, K., Soejima, H., Nakashima, N., 2022. Explanation of machine learning models using Shapley additive explanation and application for real data in hospital. Comput. Methods Prog. Biomed. 214, 106584. https://doi.org/10.1016/j.cmpb.2021.106584.

Park, S., Lee, B.K., 2013. Strong positive association of traditional Asian-style diets with blood cadmium and lead levels in the Korean adult population. Int. J. Environ. Health Res. 23, 531–543. https://doi.org/10.1080/09603123.2013.769204.

Pearce, Neil, Checkoway, Harvey, Kriebel, David, 2007. Bias in occupational epidemiology studies. OEM 64 (8), 562–568. https://www.jstor.org/stable/27732981.

Polamuri, S., 2017. How the Random Forest Algorithm Works in Machine Learning (Retrieved February 1 2025).

Qian, Y., Zhou, W., Yan, J., Li, W., Han, L., 2014. Comparing machine learning classifiers for object-based land cover classification using very high resolution imagery. Remote Sens. 7, 153–168. https://doi.org/10.3390/rs70100153.

Shapley, L.S., 1953. A value for n-person games. In: Kuhn, H.W., Tucker, A.W. (Eds.), Contributions to the Theory of Games. Princeton University Press, pp. 307–317. https://doi.org/10.1515/9781400881970-018.

Stafoggia, M., Breitner, S., Hampel, R., Basagaña, X., 2017. Statistical approaches to address multi-pollutant mixtures and multiple exposures: the state of the science. Curr. Environ. Health Rep. 4, 481–490. https://doi.org/10.1007/s40572-017-0162-z.

Wright, M.N., Ziegler, A., 2015. ranger: A Fast Implementation of Random Forests for High Dimensional Data in C++ and R. arXiv Preprint arXiv:1508.04409. https://doi.org/10.48550/arXiv.1508.04409.

Xu, H., Mao, Y., Xu, B., Hu, Y., 2021. Low-level environmental lead and cadmium exposures and dyslipidemia in adults: findings from the NHANES 2005–2016. J. Trace Elem. Med. Biol. 63, 126651. https://doi.org/10.1016/j.jtemb.2020.126651.

Yaseen, Z.M., 2021. An insight into machine learning models era in simulating soil, water bodies and adsorption heavy metals: review, challenges and solutions. Chemosphere 277, 130126. https://doi.org/10.1016/j.chemosphere.2021.130126.

Zhang, H.H., Chen, J.J., Zhu, L., Zhu, L., Zhu, R.B., 2011. Spatial patterns and variation of soil cadmium in Guangdong Province, China. J. Geochem. Explor. 109, 86–91. https://doi.org/10.1016/j.gexplo.2010.10.014.